

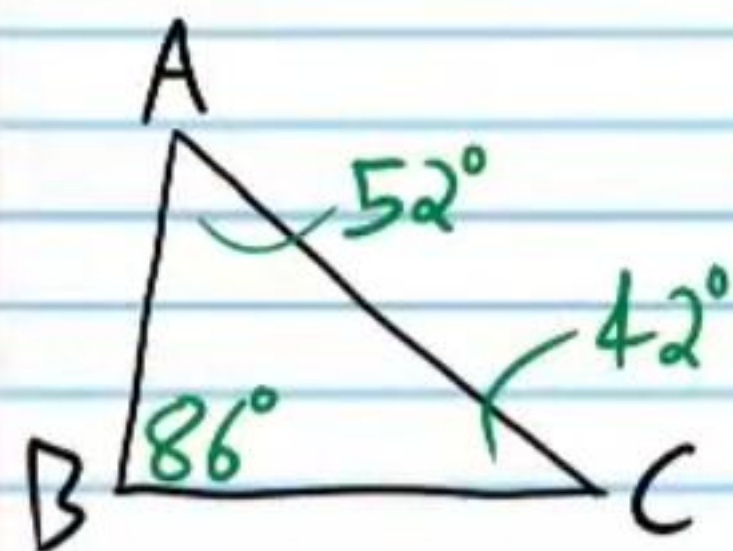
Geometry Test III

Name
Date
Course

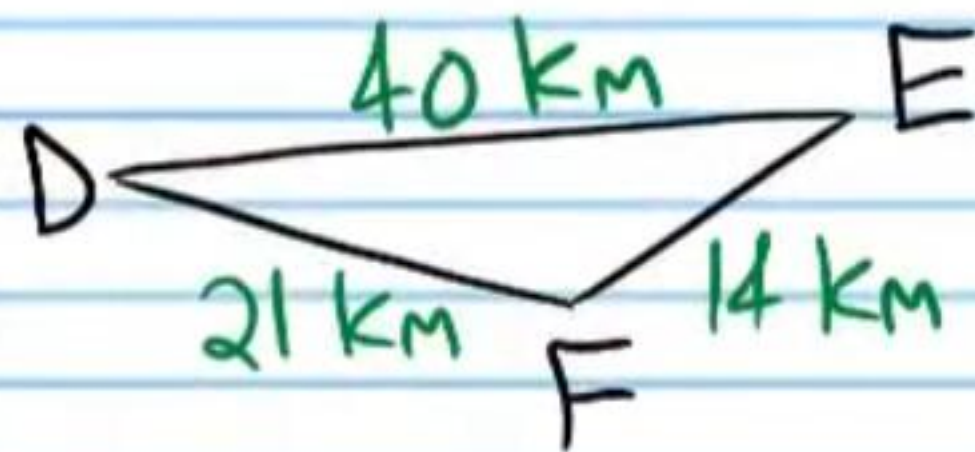
Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.
- V) A calculator may be useful for some problems on this test.

- ① List the sides of the triangle from shortest to longest.



- ② List the angles of the triangle from least to greatest.



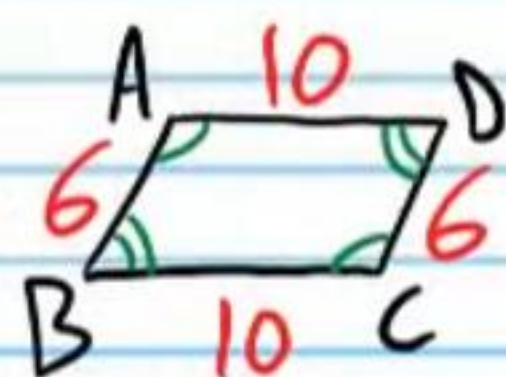
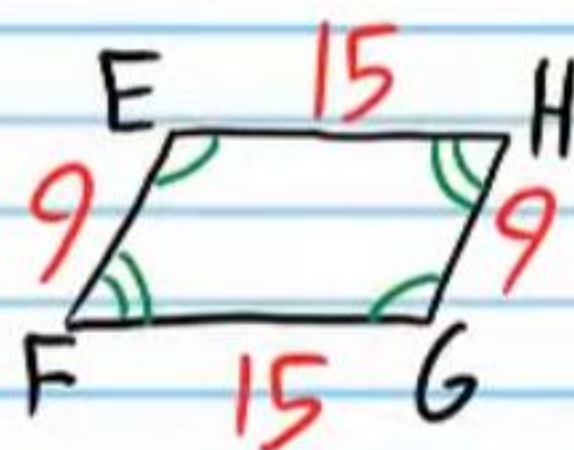
- ③ Is it possible to make a triangle with the group of side lengths below? If not, write the inequality that shows that it is not possible.

8, 6, 4

- ④ If the pair of numbers below represents side lengths of a triangle, write an inequality that shows the possible lengths of the third side x .

6, 1

Use the polygons below to answer problem #'s 5-7.



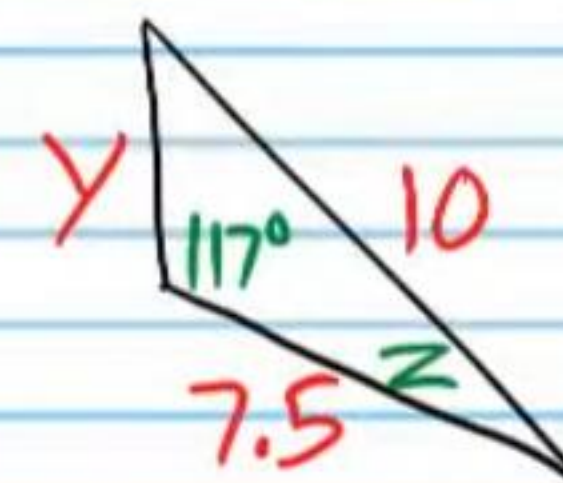
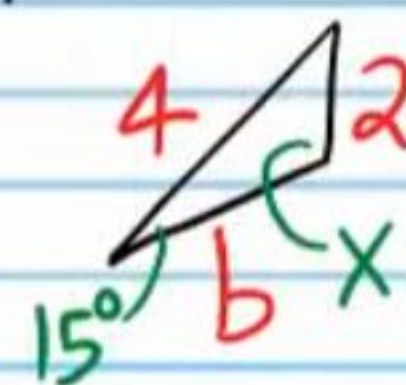
⑤ Show that the polygons are similar.

⑥ Write a similarity statement:
Write in alphabetical order.

⑦ i) Write the scale factor:

ii) Write the scale factor in a different way
(Hint: Remember, there are two ways to write a scale factor).

The polygons below are similar. Use them to answer problem #'s 8-9.



⑧ i) Find x .

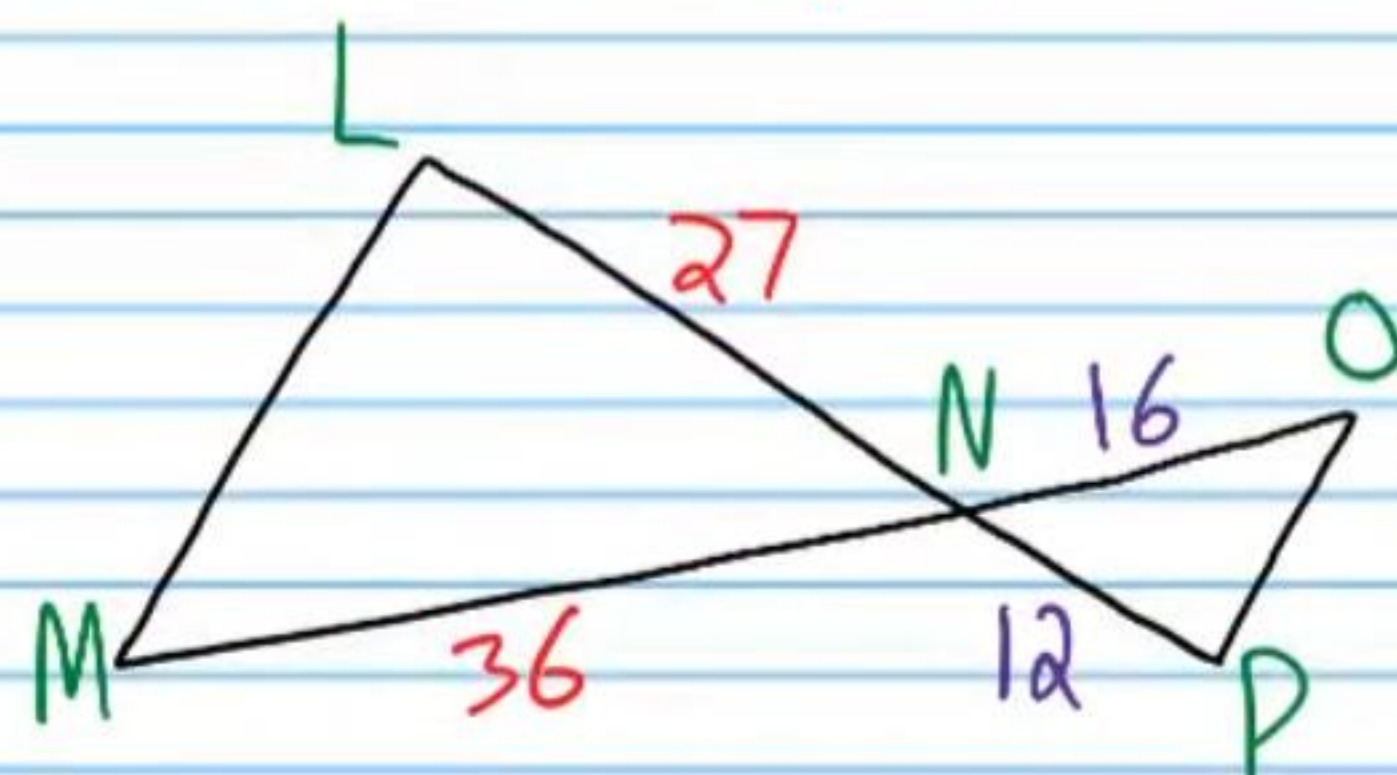
ii) Find y .

⑨ i) Find z

ii) Find b .

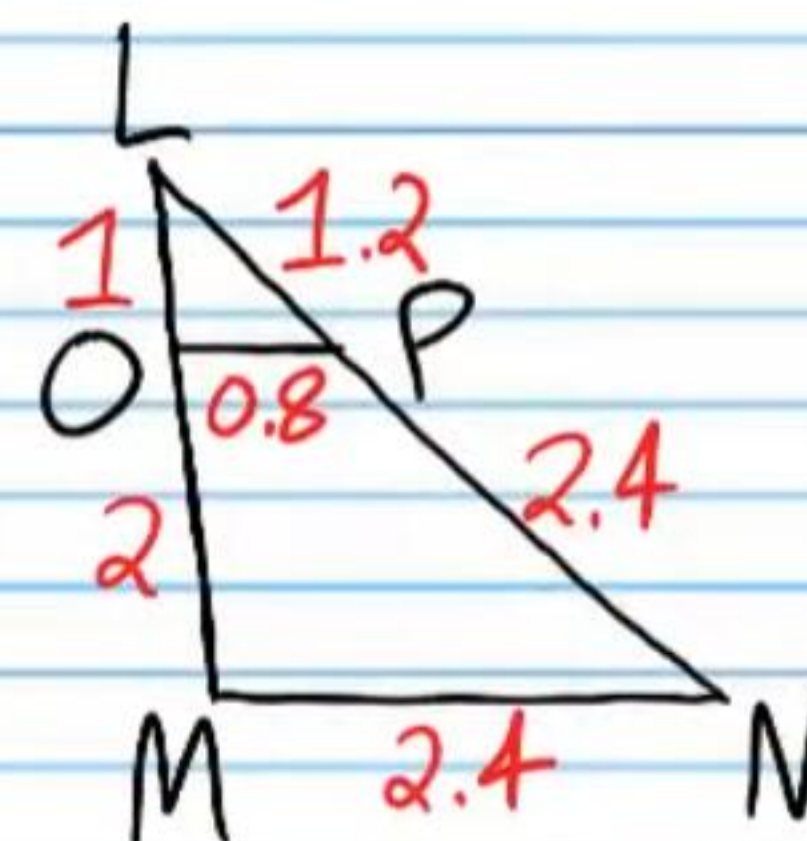
In the following problems, write two-column proofs showing that the triangles are similar. You might not use the same number of steps written in Roman numerals. You may use more or less.

⑩



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	

⑪



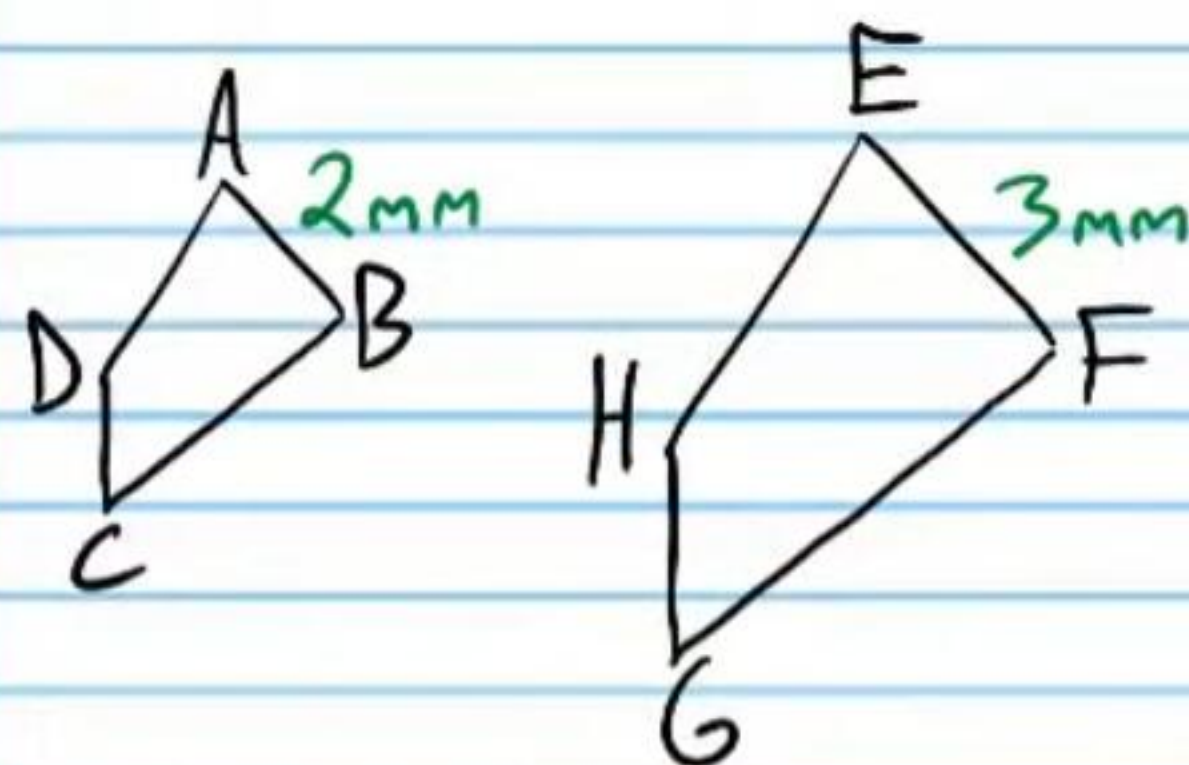
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	

⑫ Find x .

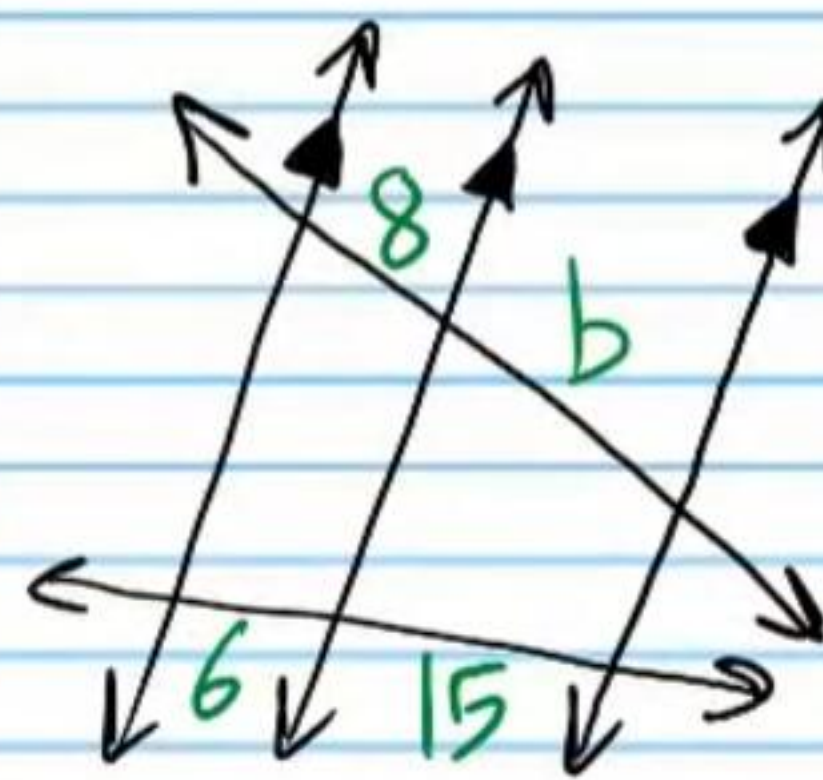


⑬ Find the perimeter of ABCD if the polygons below are similar.

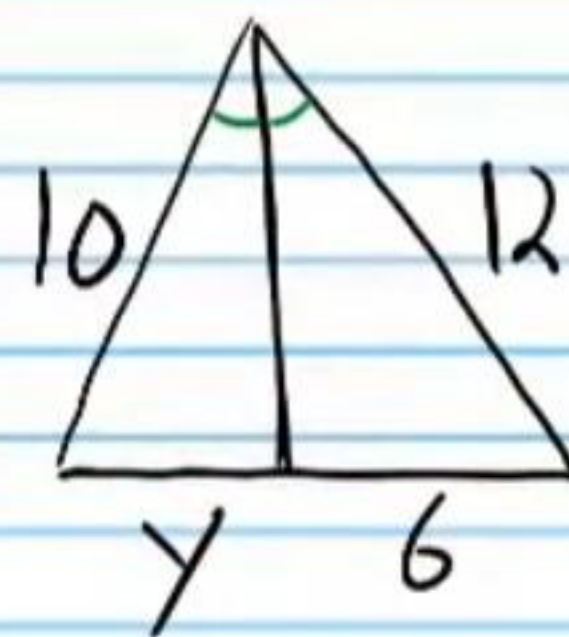
$$P_1 = ? \quad P_2 = 16 \text{ mm}$$



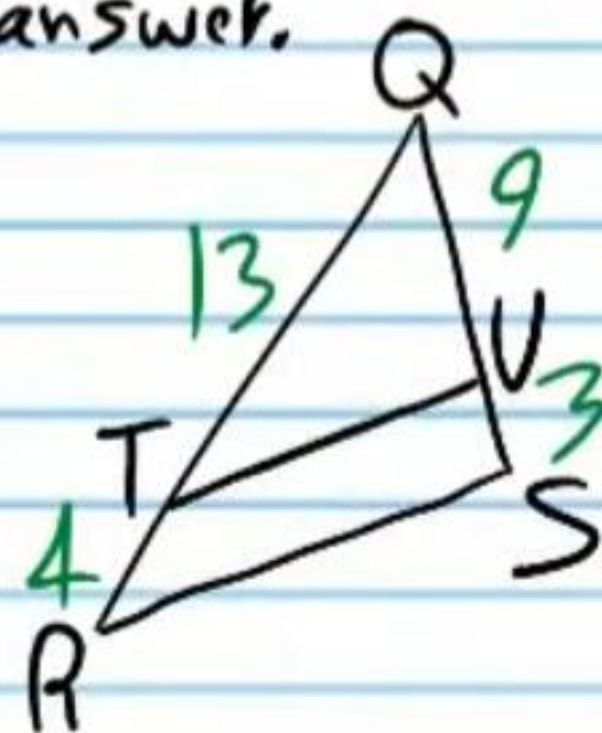
⑭ Find b .



⑮ Find y .

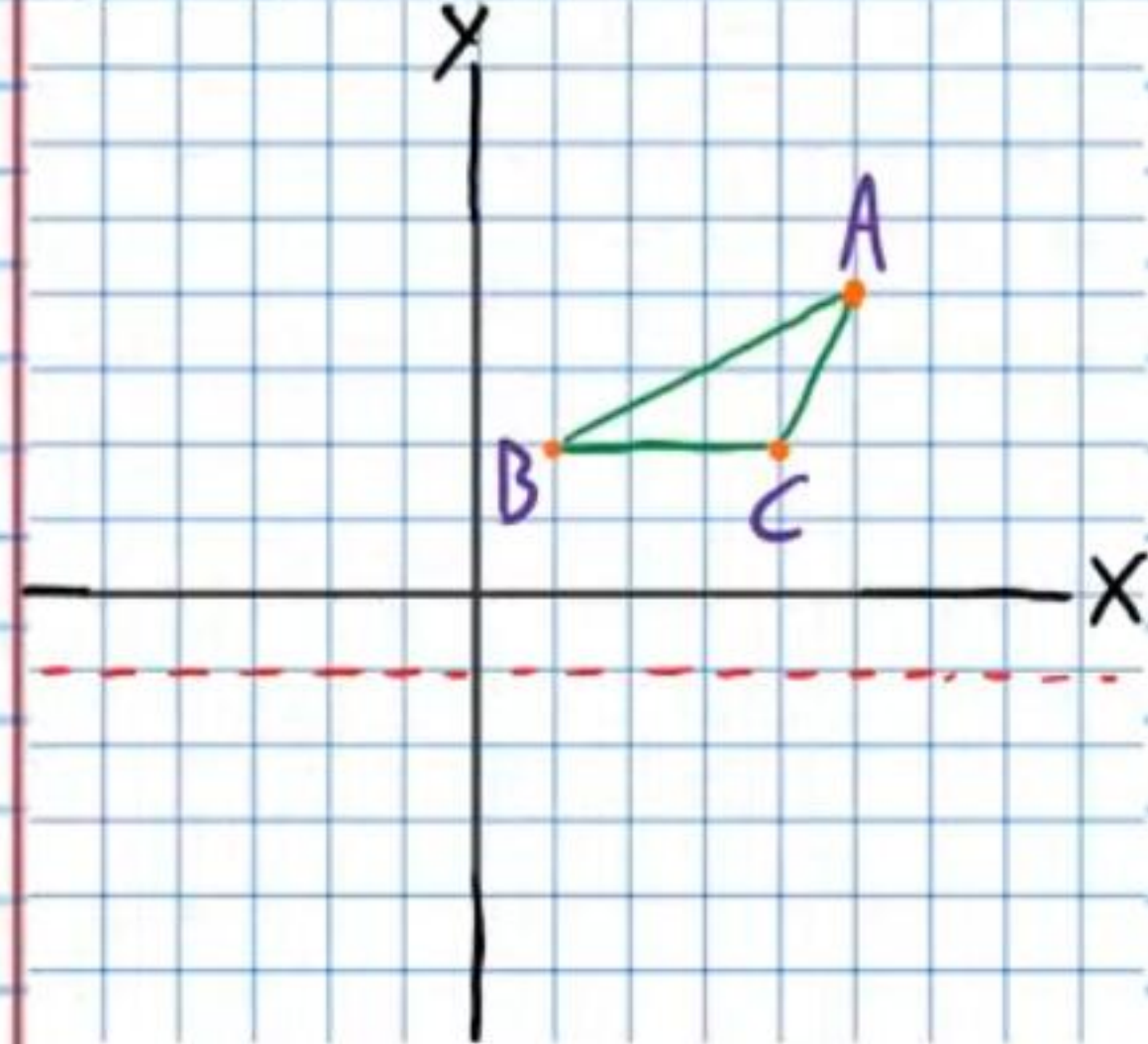


⑯ Is $\overline{TU} \parallel \overline{RS}$? Show the proportions that prove your answer.

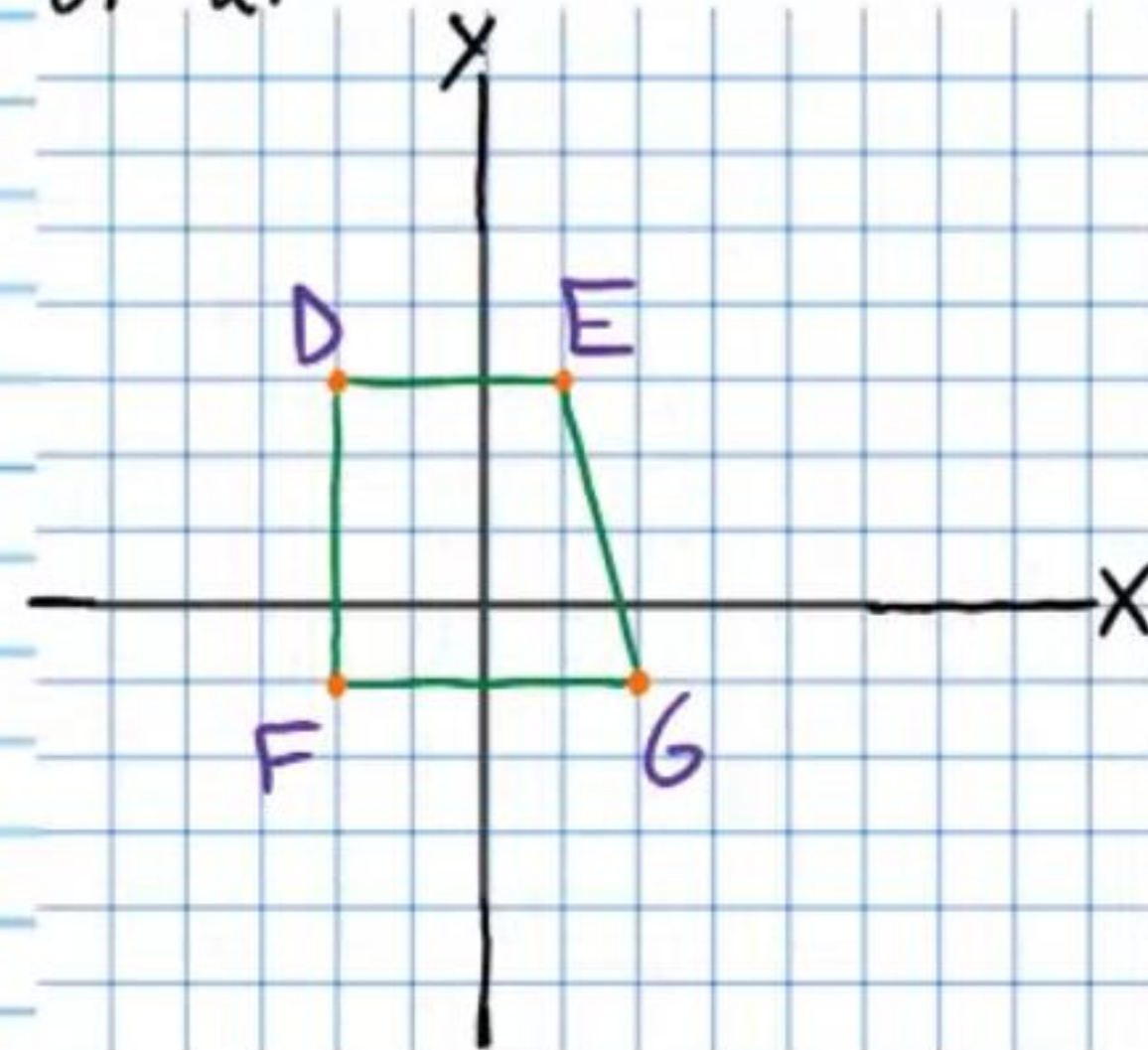


Given the "pre-image," graph the "image" using the given transformation.

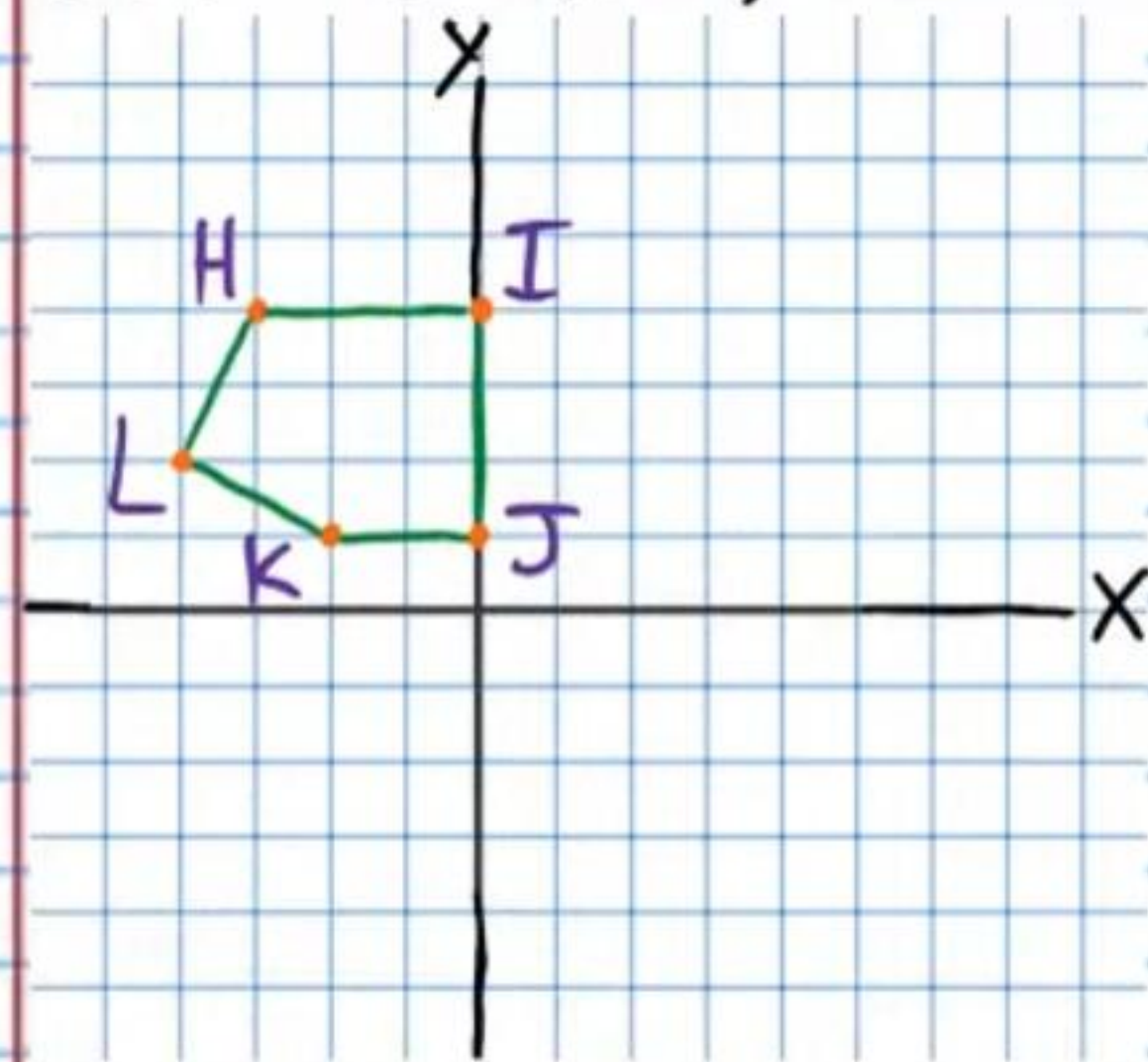
(17) Reflection about the dotted line.



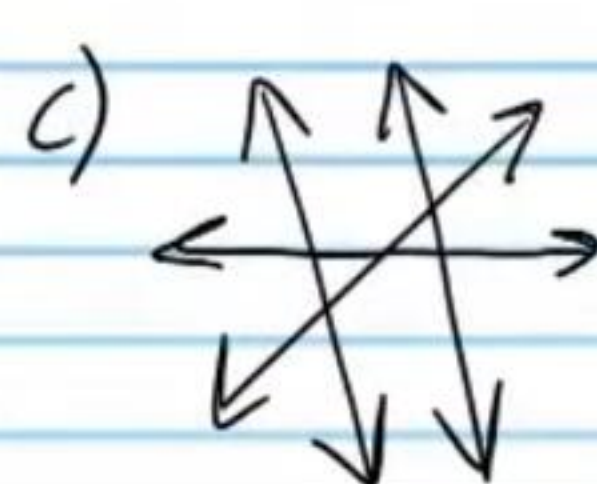
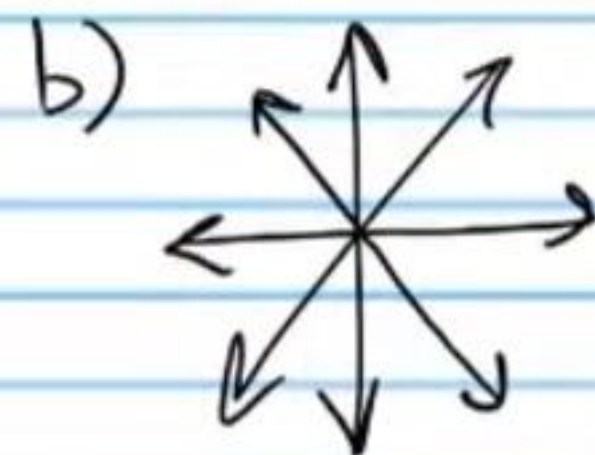
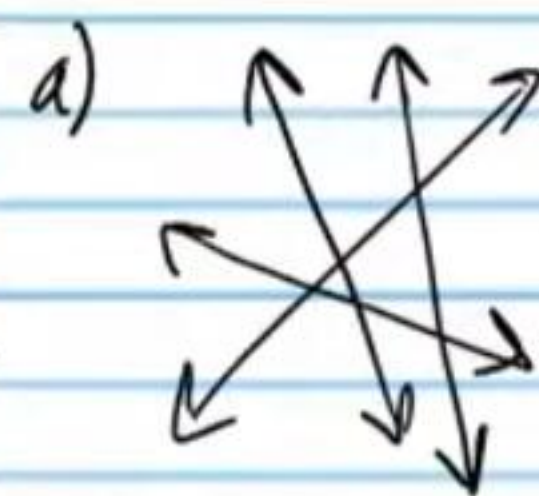
(18) Dilation about the origin with a scale factor of 2.



(19) Translation:
 $(x, y) \rightarrow (x+2, y-2)$

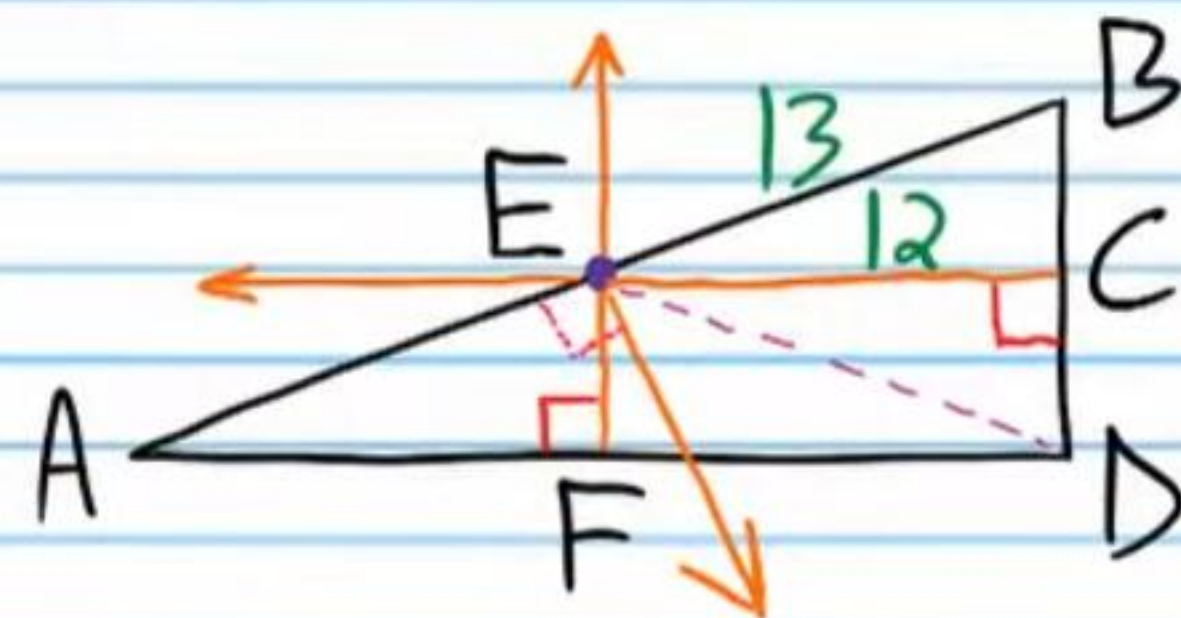


(20) Circle each group that contains concurrent lines.
You may need to circle more than one group.



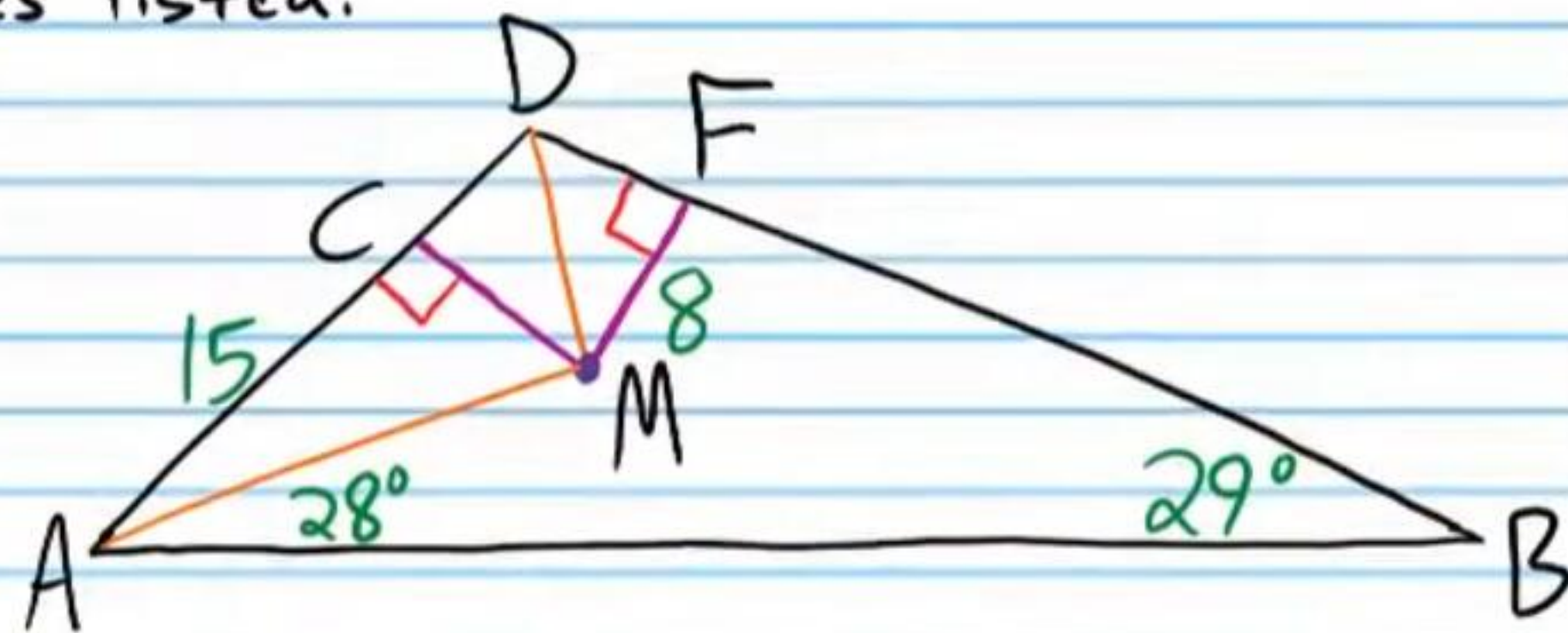
(21) If point E is the circumcenter of the triangle below, find the quantities listed.

- i) AE
- ii) DE
- iii) BC



iv) CD

②② If point M is the incenter of the triangle below, find the quantities listed.



i) $m\angle CAM$

ii) $m\angle MDF$

iii) $m\angle CDF$

iv) AM

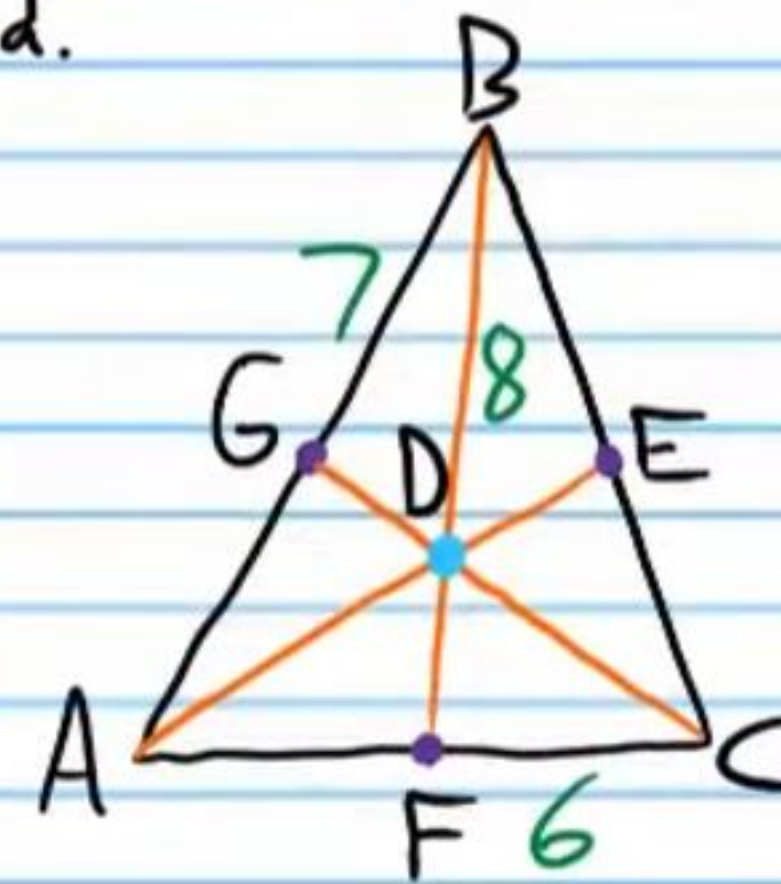
②③ If the segments below are the medians of the triangle, find the quantities listed.

i) DF

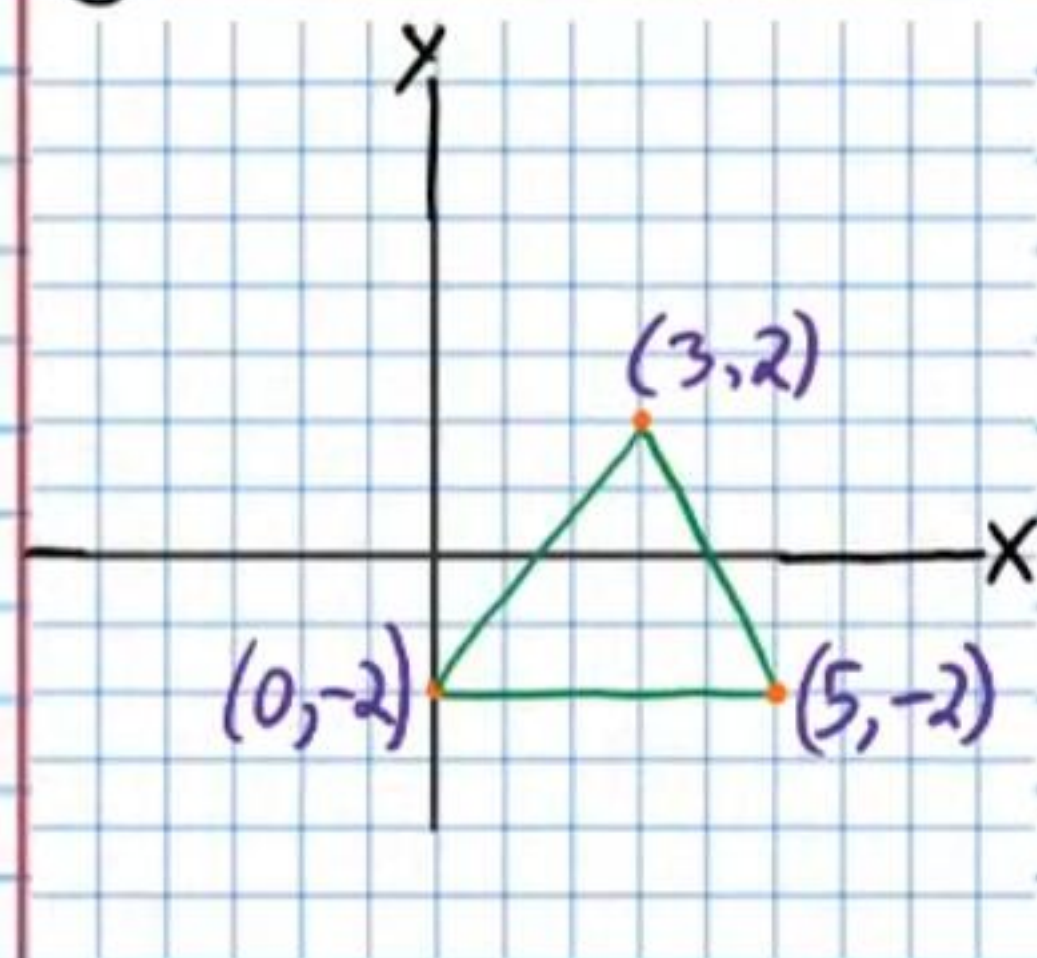
ii) AC

iii) GA

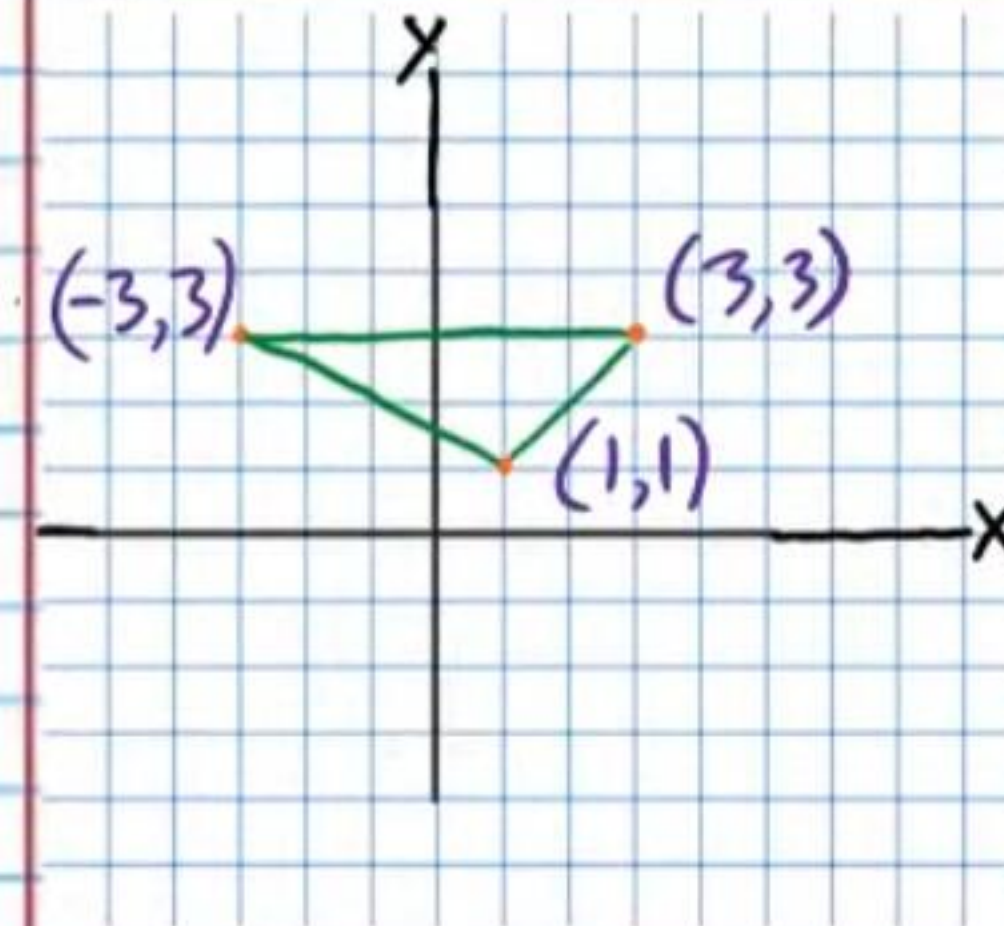
iv) If $GC = 9$, find DC .



(24) Find the coordinates of the orthocenter of the triangle below.



(25) Find the coordinates of the centroid of the triangle below.



Geometry Test III Answers

① AB, BC, AC

② $\angle D, \angle E, \angle F$

③ Yes

④ $5 < x < 7$

⑤ I) $\angle E \cong \angle A, \angle H \cong \angle D, \angle F \cong \angle B, \angle G \cong \angle C$.

$$\text{II) } \frac{9}{6} = \frac{9}{6} = \frac{15}{10} = \frac{15}{10}$$

$$\frac{3}{2} = \frac{3}{2} = \frac{3}{2} = \frac{3}{2}$$

$$\frac{EF}{AB} = \frac{HG}{DC} = \frac{EH}{AD} = \frac{FG}{BC}$$

⑥ $\square ABCD \cong \square EFGH$

⑦ i) $\frac{3}{2}$ or $\frac{2}{3}$

ii) If you wrote $\frac{3}{2}$ for part i, then $\frac{2}{3}$ is the answer for this part. If you wrote $\frac{2}{3}$ for part i, then $\frac{3}{2}$ is the answer for this part.

⑧ i) $x = 117^\circ$ ii) $y = 5$

⑨ i) $z = 15^\circ$ ii) $b = 3$

⑩

Statement	Reason
I) $\angle LNM \cong \angle PNO$	Vert. $\angle$'s Thm.
II) $LN = 27, NP = 12, MN = 36,$ and $NO = 16$.	Given.
III) $\frac{LN}{NP} = \frac{27}{12}$ and $\frac{MN}{NO} = \frac{36}{16}$	Subs. Prop of =.
IV) $\frac{LN}{NP} = \frac{9}{4}$ and $\frac{MN}{NO} = \frac{9}{4}$	Simplification.
V) $\frac{LN}{NP} = \frac{MN}{NO}$	Trans. Prop of =.
VI) $\triangle MLN \sim \triangle OPN$	SAS

⑪

Statement	Reason
I) $LO = 1, LP = 1.2, MO = 2,$ $PN = 2.4, OP = 0.8, \& MN = 2.4$.	Given (diagram)
II) $LM = LO + OM \& LN = LP + PN$	Seg Add. Post.
III) $LM = 1 + 2 \& LN = 1.2 + 2.4$	Subs. Prop of =.
IV) $LM = 3 \& LN = 3.6$	Simplification.
V) $\frac{LO}{LM} = \frac{1}{3}, \frac{LP}{LN} = \frac{1.2}{3.6}, \& \frac{OP}{MN} = \frac{0.8}{2.4}$	Subs. Prop of =.

$$\text{VI) } \frac{LP}{LN} = \frac{1}{3} \text{ \& } \frac{OP}{MN} = \frac{1}{3}$$

$$\text{VII) } \frac{LO}{LM} = \frac{LP}{LN} = \frac{OP}{MN}$$

$$\text{VIII) } \triangle LOP \sim \triangle LMN$$

Simplification.

Trans. Prop of =.

SSS

$$\textcircled{12} \ x = 6$$

$$\textcircled{13} \ P_1 = 10\frac{2}{3} \text{ mm or } \frac{32}{3} \text{ mm or } 10.\bar{6} \text{ mm}$$

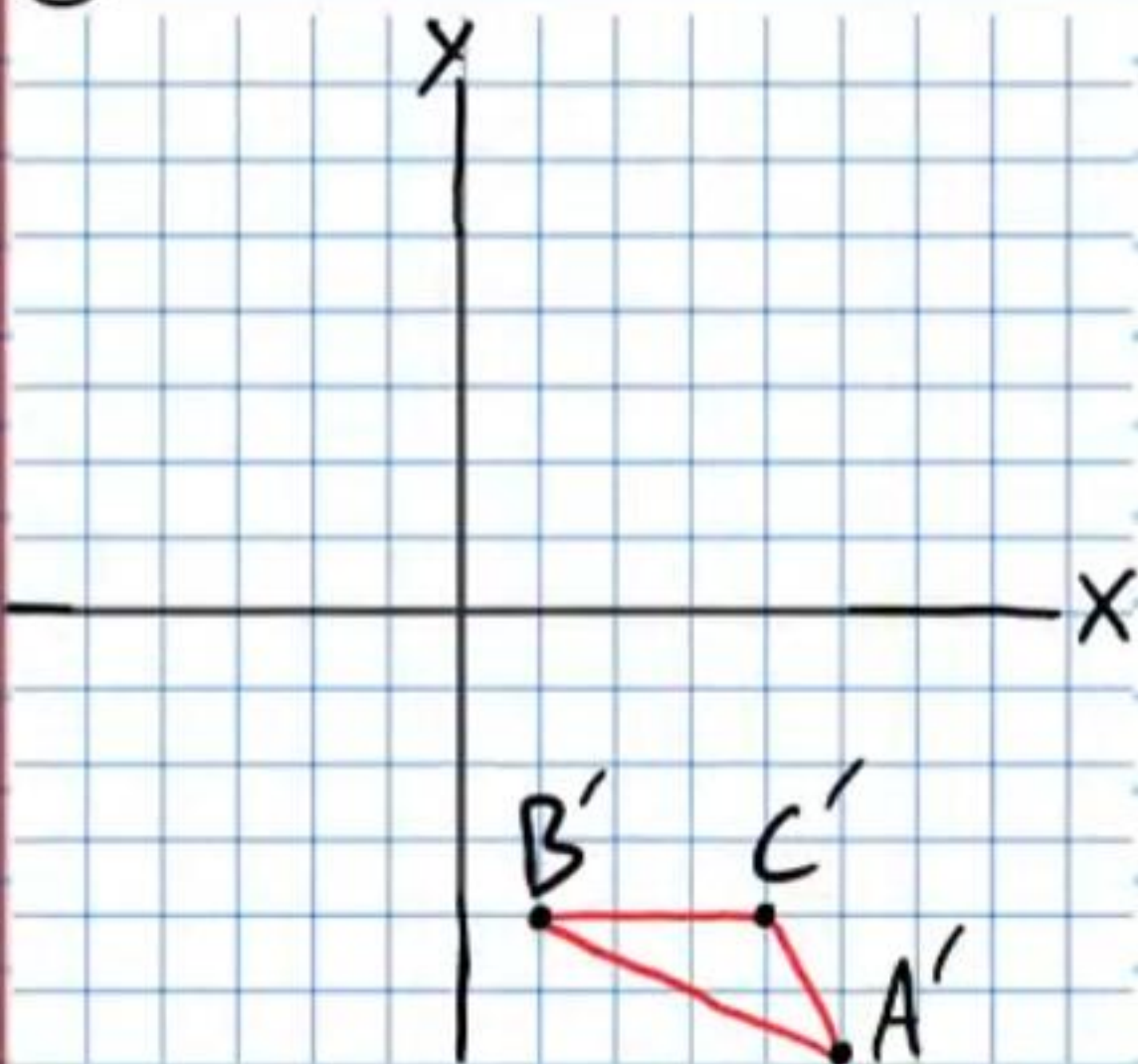
$$\textcircled{14} \ b = 20$$

$$\textcircled{15} \ y = 5$$

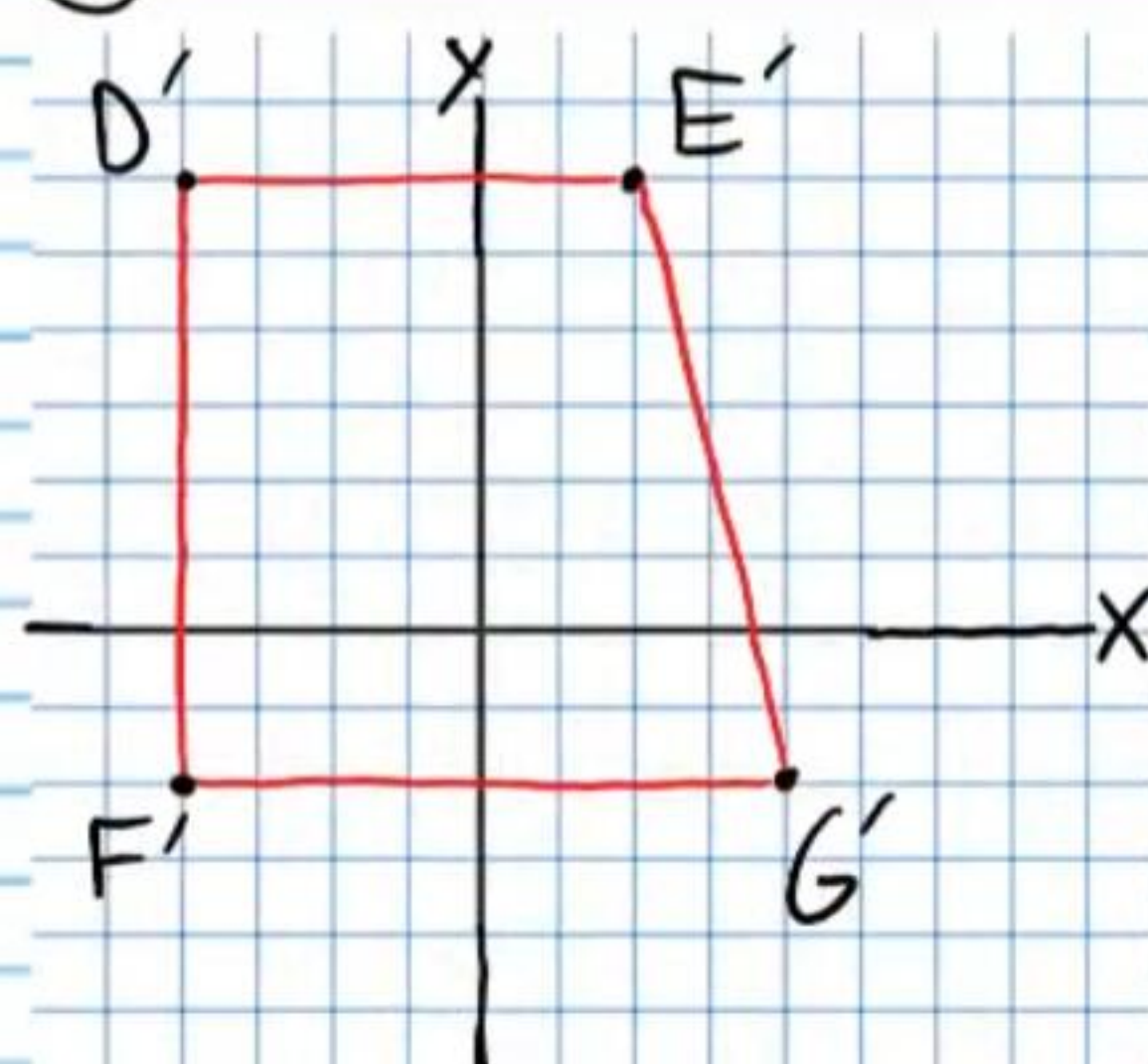
$$\textcircled{16} \ \text{No} \quad \frac{13}{4} \neq \frac{9}{3}$$

$$13/4 \neq 3$$

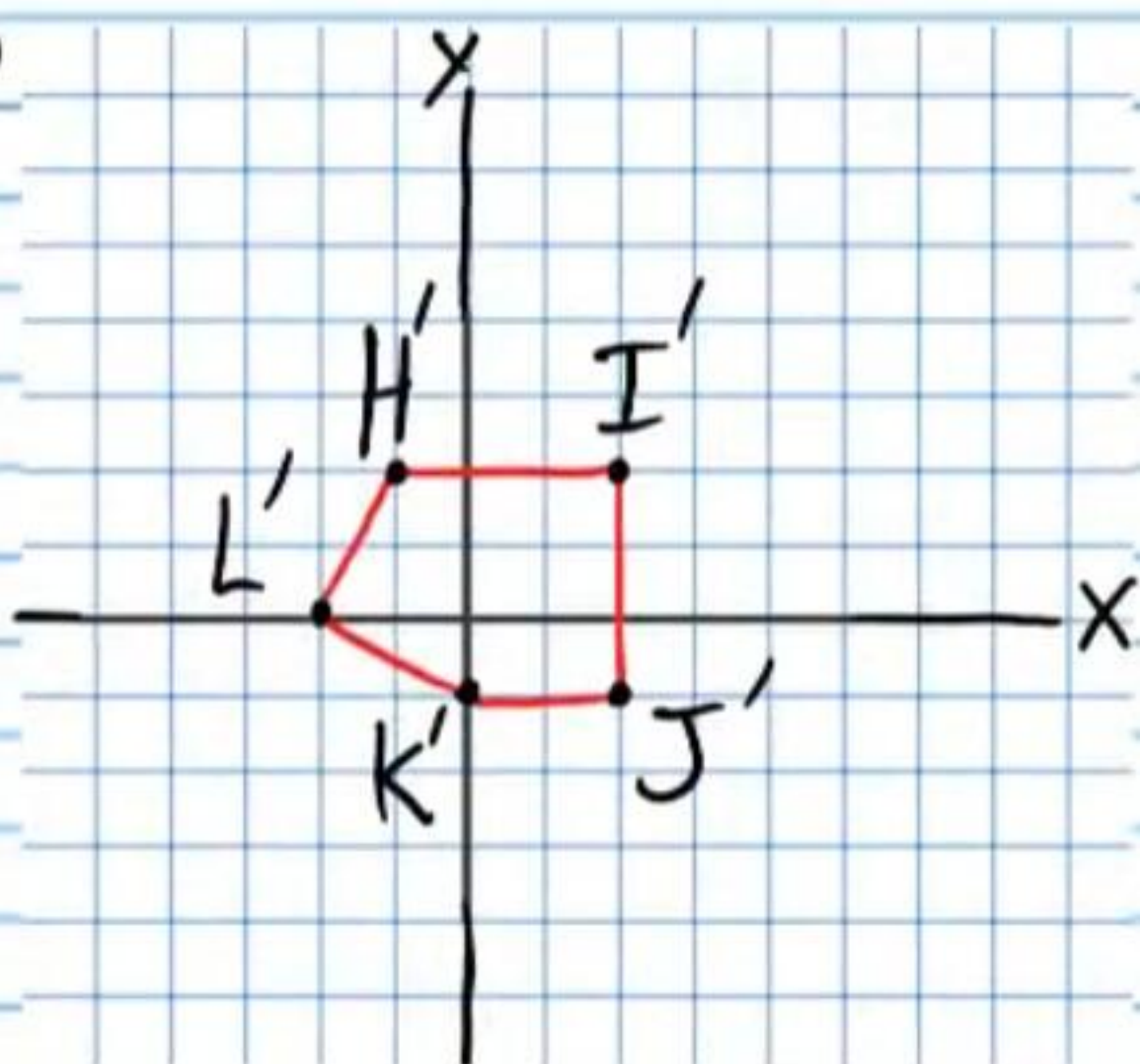
$\textcircled{17}$



$\textcircled{18}$



$\textcircled{19}$



$$\textcircled{20} \ b \ \& \ d$$

$$\textcircled{21} \ \text{i) } AE = 13 \quad \text{ii) } DE = 13 \quad \text{iii) } BC = 5 \quad \text{iv) } CD = 5$$

$$\textcircled{22} \ \text{i) } m\angle CAM = 28^\circ \quad \text{ii) } m\angle MDF = 47.5^\circ$$

$$\text{iii) } m\angle CDF = 95^\circ \quad \text{iv) } AM = 17$$

$$\textcircled{23} \ \text{i) } DF = 4 \quad \text{ii) } AC = 12 \quad \text{iii) } GA = 7 \quad \text{iv) } DC = 6$$

$$\textcircled{24} \ (3, -\frac{1}{2})$$

$$\textcircled{25} \ (\frac{1}{3}, 2\frac{1}{3}) \text{ or } (\frac{1}{3}, \frac{7}{3})$$